

Besmate Inhalation Solution Unit Dose PE Bottles 2.5ml

【PRODUCT DESCRIPTION】

It is an almost colorless, clear liquid.

【INGREDIENTS】

Each ml contains:

Ipratropium Bromide B.P. 0.2 mg
Salbutamol (as Sulfate) B. P. 1.0 mg

【PHARMACODYNAMICS】

Ipratropium bromide is a quaternary ammonium compound with anticholinergic (parasympatholytic) properties. In nonclinical studies, it appears to inhibit vagally mediated reflexes by antagonizing the action of acetylcholine, the transmitter agent released from the vagus nerve. Anticholinergics prevent the increase in intracellular concentration of Ca^{++} which is caused by interaction of acetylcholine with the muscarinic receptor on bronchial smooth muscle. Ca^{++} release is mediated by the second messenger system consisting of IP3 (inositol triphosphate) and DAG (diacylglycerol).

The bronchodilation following inhalation of ipratropium bromide is primarily local and site specific to the lung and not systemic in nature. Salbutamol sulfate is a β_2 -adrenergic agent which acts on airway smooth muscle resulting in relaxation. Salbutamol relaxes all smooth muscles from the trachea to the terminal bronchioles and protects against all bronchoconstrictor challenges.

Ipratropium Bromide & Salbutamol Inhalation Solution provides the simultaneous release of ipratropium bromide and salbutamol sulfate allowing the additive effect on both muscarinic and β_2 -adrenergic receptors in the lung resulting in a bronchodilation which is superior to that provided by each single agent.

【PHARMACOKINETICS】

Ipratropium

Cumulative renal excretion (0-24 hrs) of ipratropium (parent compound) is approximated to 46% of an intravenously administered dose, below 1% of an oral dose and approximately 3-4% of an inhaled dose. Based on these data, the total systemic bioavailability of oral and inhaled doses of ipratropium bromide is estimated at 2% and 7 to 9% respectively. Taking this into account, swallowed dose portions of ipratropium bromide do not relevantly contribute to systemic exposure.

Kinetic parameters describing the disposition of ipratropium were calculated from plasma concentrations after i.v. administration. A rapid biphasic decline in plasma concentrations is observed. The apparent volume of distribution at steady-state (V_{dss}) is approximately 176 L (≈ 2.4 L/kg). The drug is minimally (less than 20%) bound to plasma proteins. Preclinical studies with rats and dogs revealed that the quaternary amine ipratropium does not cross the blood-brain barrier. The half-life of the terminal elimination phase is approximately 1.6 hours. Ipratropium has a total clearance of 2.3 L/min and a renal clearance of 0.9 L/min. After intravenous administration approximately 60% of a dose is metabolized probably mainly in the liver by oxidation.

In an excretion balance study cumulative renal excretion (6 days) of drug-related radioactivity (including parent compound and all metabolites) accounted for 72.1% after intravenous administration, 9.3% after oral administration and 3.2% after inhalation. Total radioactivity excreted via the faeces was 6.3% following intravenous application. 88.5% following oral dosing and 69.4% after inhalation. Regarding the excretion of drug-related radioactivity after intravenous administration, the main excretion occurs via the kidneys. The half-life for elimination of drug-related radioactivity (parent compound and metabolites) is 3.6 hours. The main urinary metabolites bind poorly to the muscarinic receptor and have to be regarded as ineffective.

Salbutamol

Salbutamol is rapidly and completely absorbed following oral administration either by the inhaled or gastric route and has an oral bioavailability of approximately 50%. Mean peak plasma salbutamol concentrations of 492 pg/ml occur within three hours after inhalation of Ipratropium Bromide & Salbutamol Inhalation Solution. Following this single inhaled administration, approximately 27% of the estimated mouthpiece dose is excreted unchanged in the 24-hour urine. Kinetic parameters were calculated from plasma concentrations after i.v. administration. The apparent volume of distribution (V_z) is approximately 156 L (≈ 2.5 L/kg). Only 8% of the drug is bound to plasma proteins.

Salbutamol will cross the blood brain barrier reaching concentrations amounting to about 5% of the plasma concentrations. The mean terminal half-life is approximately 4 hours with a mean total clearance of 480 mL/min and a mean

renal clearance of 291 mL/min. Salbutamol is conjugatively metabolized to salbutamol 4'-O-sulfate. The R(-)-enantiomer of salbutamol (levosalbutamol) is preferentially metabolized and is therefore cleared from the body more rapidly than the S(+)-enantiomer. Following intravenous administration, urinary excretion was complete after approximately 24 hours. The majority of the dose was excreted as parent compound (64.2%) and 12.0% were excreted as sulfate conjugate. After oral administration urinary excretion of unchanged drug and sulfate conjugate were 31.8% and 48.2% of the dose, respectively.

Co-administration of ipratropium bromide and salbutamol sulfate does not potentiate the systemic absorption of either component and that therefore the additive activity of Ipratropium Bromide & Salbutamol Inhalation Solution is due to the combined local effect on the lung following inhalation.

【INDICATIONS】

For the management of reversible bronchospasm associated with obstructive airway diseases in patients who require more than a single bronchodilator.

【RECOMMENDED DOSE】

Ipratropium Bromide and Salbutamol inhalation solution should be restricted to doctors' use only.

Besmate Inhalation Solution in unit dose bottles may be administered from a suitable nebuliser or an intermittent positive pressure ventilator.

The following dose is recommended for adults (Including Elderly Patients) and adolescents >12 years:

Treatment of Acute Attacks:

1 unit dose bottle is sufficient for prompt symptom relief in many cases. In severe cases if an attack has not been relieved by one unit dose bottle, the administration of a second unit dose bottles may be required. In these cases, patients should consult the doctor or the nearest hospital immediately.

Maintenance Treatment:

1 unit dose bottle three or four times daily.

Instructions for use:

The unit dose bottles are intended only for inhalation with suitable nebulising devices and should not be taken orally or administered parenterally.

1. Prepare the nebuliser for filling, according to the instructions provided by the manufacturer or doctor.
2. Open the unit dose bottle by firmly twisting the top.
3. Squeeze the content of the unit dose bottle into the nebuliser reservoir.
4. Assemble the nebuliser and use as directed.
5. After use throw away any solution left in the reservoir and clean the nebuliser, following the manufacturer's instructions.

Since the unit dose bottles contain no preservative, it is important that the contents are used soon after opening and that a fresh bottle is used for each administration to avoid microbial contamination. Partly used, opened or damaged unit dose bottles should be discarded.

It is strongly recommended not to mix Ipratropium Bromide & Salbutamol Inhalation Solution with other drugs in the same nebuliser.

【ROUTE OF ADMINISTRATION】

Respiratory (Inhalation)

【WARNING AND PRECAUTIONS】

Immediate hypersensitivity reactions may occur after administration of Ipratropium Bromide & Salbutamol Inhalation Solution, as demonstrated by rare cases of urticaria, angioedema, rash, bronchospasm and oropharyngeal oedema.

There have been isolated reports of ocular complications (ie, mydriasis, increased intraocular pressure, narrow-angle glaucoma, eye pain) when aerosolised ipratropium bromide either alone or in combination with an adrenergic β_2 -agonist, has escaped into the eyes.

Eye pain or discomfort, blurred vision, visual halos or colored images in association with red eyes from conjunctival congestion and corneal oedema may be signs of acute narrow-angle glaucoma. Should any combination of these symptoms develop, treatment with miotic drops should be initiated and specialist advice sought immediately.

Patients must be instructed in the correct administration of Ipratropium Bromide & Salbutamol Inhalation Solution. Care must be taken not to allow the solution or mist to enter into the eyes. Patients who may be predisposed to glaucoma should

be warned specifically to protect their eyes.

It is recommended that the nebulised Ipratropium Bromide & Salbutamol Inhalation Solution be administered via a mouth piece. If this is not available and a nebuliser mask is used, it must fit properly.

In the following conditions, Ipratropium Bromide & Salbutamol Inhalation Solution should only be used after careful risk/benefit assessment, especially when doses higher than recommended are used: insufficiently controlled diabetes mellitus, recent myocardial infarction, severe organic heart or vascular disorders, hyperthyroidism, phaeochromocytoma, risk of narrow-angle glaucoma, prostatic hypertrophy or bladder-neck obstruction.

Cardiovascular effects may be seen with sympathomimetic drugs, including Ipratropium Bromide & Salbutamol Inhalation Solution. There is some evidence from post-marketing data and published literature of rare occurrences of myocardial ischaemia associated with salbutamol. Patients with underlying severe heart disease (eg, ischaemic heart disease, tachyarrhythmia or severe heart failure) who are receiving salbutamol for respiratory disease, should be warned to seek medical advice if they experience chest pain or other symptoms of worsening heart disease. Attention should be paid to assessment of symptoms as dyspnoea and chest pain, as they may be of either respiratory or cardiac origin.

Potentially serious hypokalaemia may result from β_2 -agonist therapy. Additionally, hypoxia may aggravate the effects of hypokalaemia on cardiac rhythm.

Patients with cystic fibrosis may be more prone to gastrointestinal motility disturbances.

In the case of acute, rapidly worsening dyspnoea (difficulty in breathing) a physician should be consulted immediately.

If higher than recommended doses of Ipratropium Bromide & Salbutamol Inhalation Solution are required to control symptoms, the patients' therapy plan should be reviewed by a physician.

The use of Ipratropium Bromide & Salbutamol Inhalation Solution may lead to positive results with regards to salbutamol in tests for nonclinical substance abuse eg, in the context of athletic performance enhancement (doping).

【CONTRAINDICATIONS】

Hypertrophic obstructive cardiomyopathy, tachyarrhythmia.

Hypersensitivity to any of the components of the product, to atropine or its derivatives.

【SIDE EFFECTS】

Immune System Disorders:

Anaphylactic reaction
Hypersensitivity

Metabolism and Nutrition Disorders:

Hypokalaemia

Psychiatric Disorders:

Nervousness
Mental disorder

Nervous System Disorders:

Headache
Tremor
Dizziness

Eye Disorders:

Accommodation disorder
Corneal oedema
Glaucoma
Intraocular pressure increased
Mydriasis
Vision blurred
Eye pain
Conjunctival hyperaemia
Halo vision

Cardiac Disorders:

Palpitations
Tachycardia
Arrhythmia
Atrial fibrillation
Supraventricular tachycardia
Myocardial ischaemia

Respiratory, Thoracic and Mediastinal Disorders:

Cough
Dysphonia
Dry throat
Bronchospasm

Bronchospasm paradoxical
Laryngospasm
Pharyngeal oedema

Gastrointestinal Disorders:

Dry mouth
Nausea
Throat irritation
Diarrhoea
Vomiting
Constipation
Gastrointestinal motility disorder
Oedema mouth
Stomatitis

Skin and Subcutaneous Tissue Disorders:

Skin reactions such as
- Rash
- Pruritus
- Urticaria
Angioedema
Hyperhidrosis

Musculoskeletal and Connective Tissue Disorders:

Muscle spasms
Muscular weakness
Myalgia

Renal and Urinary Disorders:

Urinary retention

General Disorders and Administration Site Conditions:

Asthenia

Investigations:

Blood pressure diastolic decreased
Blood pressure systolic increased

【PREGNANCY AND LACTATION】

The safety of Ipratropium Bromide & Salbutamol Inhalation Solution during human pregnancy is not established. The usual precautions regarding the use of drugs in pregnancy, especially during the first trimester, should be observed.

The inhibitory effect of Ipratropium Bromide & Salbutamol Inhalation Solution on uterine contraction should be taken into account.

Salbutamol sulfate and ipratropium bromide are probably excreted in breast milk and their effects on the neonate are not known. Although lipid-insoluble quaternary bases pass into breast milk, it is unlikely that ipratropium bromide would reach the infant to an important extent, especially when taken by inhalation. However, because many drugs are excreted in breast milk, caution should be exercised when Ipratropium Bromide & Salbutamol Inhalation Solution is administered to a nursing woman.

【SYMPTOMS AND TREATMENT OF OVERDOSE】

Symptoms:

The effects of overdosage are expected to be primarily related to salbutamol.

The expected symptoms with overdosage are those of excessive beta-adrenergic-stimulation, the most prominent being tachycardia, palpitation, tremor, hypertension, hypotension, hypokalaemia, widening of the pulse pressure, anginal pain, arrhythmias, and flushing. Metabolic acidosis has also been observed with overdosage of Salbutamol.

Expected symptoms of overdosage with ipratropium bromide (such as dry mouth, visual accommodation disorders) are mild and transient in nature in view of the wide therapeutic range and topical administration.

Therapy:

Treatment with Ipratropium Bromide & Salbutamol Inhalation Solution should be discontinued. Acid base and electrolyte monitoring should be considered.

Administration of sedatives, tranquilizers, in severe cases intensive therapy. Beta-receptor blockers, preferably β_{1-} selective, are suitable as specific antidotes; however, a possible increase in bronchial obstruction must be taken into account and the dose should be adjusted carefully in patients suffering from bronchial asthma.

【INTERACTIONS WITH OTHER MEDICAMENTS】

The concurrent administration of xanthine derivatives as well as other β -adrenergics and anticholinergics may increase the side effects.

Beta-agonist (beta-adrenergic agonist) induced hypokalaemia may be increased by concomitant treatment with xanthine derivatives,

glucocorticosteroids and diuretics. This should be taken into account particularly in patients with severe airway obstruction.

Hypokalaemia may result in an increased susceptibility to arrhythmias in patients receiving digoxin.

It is recommended that serum potassium levels are monitored in such situations.

A potentially serious reduction in bronchodilator effect may occur during concurrent administration of β -blockers.

Beta-adrenergic agonists should be administered with caution to patients being treated with monoamine oxidase inhibitors or tricyclic antidepressants, since the action of β -adrenergic agonists may be enhanced.

Inhalation of halogenated hydrocarbon anaesthetics such as halothane, trichloroethylene and enflurane may increase the susceptibility to the cardiovascular effects of β -agonists.

【STORAGE】

Store below 30°C, and protect from light. Do not freeze.
Keep out of reach of children.

【DOSAGE FORMS AND PACKAGING AVAILABLE】

2.5 ml / Plastic Bottle, 100 Bottles / Box

MAL No:

Manufactured by:

TAIWAN BIOTECH CO., LTD.

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Product Registration Holder:

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